

2MVAa: Multivariable and Vector Analyses 1

Solutions to Exercise Sheet 1

Qu. 1. Define what is meant by a vertical section and a level curve of a function of two variables x and y .

Let $z = f(x, y)$ where $f(x, y) = \frac{x^2 + y^2}{x}$. Sketch the vertical sections and level curves of f associated with the constants 0, 1 and 2.

Solution. A vertical section is the graph of $z = f(x, c)$ or $z = f(c, y)$ for some constant c . A level curve is the curve $c = f(x, y)$ for some constant c .

The two sections are given by $z = \frac{x^2 + c^2}{x}$ (a family of hyperbolas) and $z = \frac{c^2 + y^2}{c}$ (a family of parabolas). The level curves are $(x - c/2)^2 + y^2 = c^2/4$, i.e. a family of circles. Note that the position of the centre of the circle depends on c ; it drifts as c increases.

(Familiarity with the basics of Analytic Geometry would be sufficient to conclude that the surface in question is a cone. One can obtain the canonical form of this second-order surface, i.e. $\bar{z}^2 = \bar{x}^2/a^2 + \bar{y}^2/b^2$, by changing the orientation of the axes, i.e. by a linear transformation of the variables. In the new variables the vertical sections would be hyperbolas and parabolas would appear after intersecting the cone with an *inclined* plane. Here, i.e. in our original variables (x, y, z) the cone itself is tilted so that one of the vertical sections gives us parabolas. All this, however, is outside our course.)

Qu. 2. Assuming that the following equations define z as a function of the other variables occurring, use implicit differentiation to find $\frac{\partial z}{\partial x}$ and $\frac{\partial z}{\partial y}$ where

$$(i) \quad x^2 - y^2 + z^2 = 1 \quad (ii) \quad z^2 - 4xz + y^2 = 4 \quad (iii) \quad xy + xz + yz = 3$$

$$(iv) \quad x^2 + z^2 = 4yz \quad (v) \quad xyz = x + y + z \quad (vi) \quad x^3z^2 - z^2y^2 = \sin x.$$

Solution. (i) $x^2 - y^2 + z^2 = 1$ so differentiating both sides partially with respect to x gives $2x + 2z\frac{\partial z}{\partial x} = 0$, since $\frac{\partial z^2}{\partial x} = \frac{dz^2}{dz} \frac{\partial z}{\partial x} = 2z\frac{\partial z}{\partial x}$ by the chain rule. Hence $\frac{\partial z}{\partial x} = -x/z$, provided $z \neq 0$. Similarly $-2y + 2z\frac{\partial z}{\partial y} = 0$ so $\frac{\partial z}{\partial y} = y/z$ ($z \neq 0$).

(ii) $z^2 - 4xz + y^2 = 4$ so $2z\frac{\partial z}{\partial x} - 4z - 4x\frac{\partial z}{\partial x} = 0$ since $\frac{\partial(4xz)}{\partial x} = 4z\frac{\partial x}{\partial x} + 4x\frac{\partial z}{\partial x}$ by the product rule. Hence $\frac{\partial z}{\partial x} = 2z/(z - 2x)$, provided $z \neq 2x$. Similarly, differentiating w.r.t. y gives $2z\frac{\partial z}{\partial y} - 4x\frac{\partial z}{\partial y} + 2y = 0$ so that $\frac{\partial z}{\partial y} = -y/(z - 2x)$, again provided that $z \neq 2x$.

(iii) Differentiating $xy + xz + yz = 3$ w.r.t. x gives $y + x\frac{\partial z}{\partial x} + z + y\frac{\partial z}{\partial x} = 0$, so that

$\frac{\partial z}{\partial x} = -(y+z)/(x+y)$. Similarly (or by symmetry), $\frac{\partial z}{\partial y} = -(x+z)/(x+y)$.

(iv) $x^2 + z^2 = 4yz$ so $2x + 2z\frac{\partial z}{\partial x} = 4y\frac{\partial z}{\partial x}$ or $\frac{\partial z}{\partial x} = x/(2y-z)$, so long as $z \neq 2y$. Also, differentiating w.r.t. y we get $2z\frac{\partial z}{\partial y} = 4z + 4y\frac{\partial z}{\partial y}$ so that $\frac{\partial z}{\partial y} = -2z/(2y-z)$ (again if $z \neq 2y$).

(v) $xyz = x + y + z$ so $yz + xy\frac{\partial z}{\partial x} = 1 + \frac{\partial z}{\partial x}$, giving $\frac{\partial z}{\partial x} = (yz-1)/(1-xy)$, if $xy \neq 1$.

By symmetry, $\frac{\partial z}{\partial y} = (xz-1)/(1-xy)$, if $xy \neq 1$.

(vi) $x^3z^2 - z^2y^2 = \sin x$ so that $3x^2z^2 + x^32zz_x - 2zz_xy^2 = \cos x$, since $\frac{\partial(x^3z^2)}{\partial x} = 3x^2z^2 + x^32zz_x$ and $\frac{\partial}{\partial x}z^2y^2 = 2zz_xy^2$.

Hence, if $x^3z - zy^2 \neq 0$, $\frac{\partial z}{\partial x} = (\cos x - 3x^2z^2)/(2x^3z - y^2z)$.

Differentiating w.r.t. y gives $x^32zz_y - 2zz_yy^2 - 2yz^2 = 0$ so $\frac{\partial z}{\partial y} = zy/(x^3 - y^2)$ (if $x^3z - y^2z \neq 0$).

Qu. 3. Show that $\frac{\partial^2 f}{\partial x \partial y} = \frac{\partial^2 f}{\partial y \partial x}$ in the following cases by explicitly calculating all mixed second order derivatives of

- a) $f(x, y) = x^2y^3$,
- b) $f(x, y) = e^{xy} \cos x$,
- c) $f(x, y, z) = xe^{-z} - 3xy$.

Solution. (a) $f_x = 2xy^3$ and $f_y = 3x^2y^2$ so $f_{xy} = 6xy^2 = f_{yx}$.

(b) $f_x = ye^{xy} \cos x - e^{xy} \sin x$ and $f_y = xe^{xy} \cos x$ so $f_{xy} = e^{xy} \cos x + xye^{xy} \cos x - xe^{xy} \sin x$ and $f_{yx} = e^{xy} \cos x + xye^{xy} \cos x - xe^{xy} \sin x$.

(c) $f_x = e^{-z} - 3y$, $f_y = -3x$ and $f_z = -xe^{-z}$. Hence $f_{xy} = -3 = f_{yx}$, $f_{yz} = 0 = f_{zy}$ and $f_{zx} = -e^{-z} = f_{xz}$.

Qu. 4. Assuming that the functions are sufficiently smooth, use the chain rule to find the following:

- a) $\frac{dz}{dt}$ where $z = f(x, y)$, $x = g(t)$ and $y = h(t)$;
- b) $\frac{dz}{dt}$ where $z = f(x, y, t)$, $x = g(t)$ and $y = h(t)$;
- c) $\frac{\partial}{\partial x}f(2x + y, ye^x)$ in terms of the partial derivatives of f ;
- d) $\frac{\partial u}{\partial s}$ and $\frac{\partial u}{\partial t}$ when $u = \sqrt{x^2 + y}$, $x = e^{st}$ and $y = s + \sin t$;
- e) $\frac{\partial w}{\partial s}$ and if $w = f(x, y, z)$, where $x = g(s, t)$, $y = h(s, t)$ and $z = k(s)$;

f) $\frac{\partial u}{\partial s}$ and $\frac{\partial u}{\partial t}$ when $u = \sqrt{x^2 + y^2}$, $x = e^{st}$ and $y = s + \sin t$.

Solution. (a) $\frac{dz}{dt} = \frac{\partial f}{\partial x} \frac{dx}{dt} + \frac{\partial f}{\partial y} \frac{dy}{dt} = f_x g' + f_y h'$.

(b) $\frac{dz}{dt} = \frac{\partial f}{\partial x} \frac{dx}{dt} + \frac{\partial f}{\partial y} \frac{dy}{dt} + \frac{\partial f}{\partial t} \frac{dt}{dt} = f_x g' + f_y h' + f_t$, as $\frac{dt}{dt} = 1$.

(c) Let $u = 2x + y$ and $v = ye^x$. Then, by the Chain Rule,

$$\frac{\partial f}{\partial x} = \frac{\partial f}{\partial u} \frac{\partial u}{\partial x} + \frac{\partial f}{\partial v} \frac{\partial v}{\partial x} = 2 \frac{\partial f}{\partial u} + ye^x \frac{\partial f}{\partial v}.$$

(d) By the Chain Rule

$$\begin{aligned} \frac{\partial u}{\partial s} &= \frac{\partial u}{\partial x} \frac{\partial x}{\partial s} + \frac{\partial u}{\partial y} \frac{\partial y}{\partial s} = \frac{x}{\sqrt{x^2 + y^2}} te^{st} + \frac{1}{2\sqrt{x^2 + y^2}} 1 = \frac{2te^{2st} + 1}{2\sqrt{e^{2st} + (s + \sin t)^2}}. \\ \frac{\partial u}{\partial t} &= \frac{\partial u}{\partial x} \frac{\partial x}{\partial t} + \frac{\partial u}{\partial y} \frac{\partial y}{\partial t} = \frac{x}{\sqrt{x^2 + y^2}} se^{st} + \frac{1}{2\sqrt{x^2 + y^2}} \cos t = \frac{2se^{2st} + \cos t}{2\sqrt{e^{2st} + (s + \sin t)^2}}. \end{aligned}$$

(e) By the Chain Rule,

$$\frac{\partial w}{\partial s} = \frac{\partial f}{\partial x} \frac{\partial x}{\partial s} + \frac{\partial f}{\partial y} \frac{\partial y}{\partial s} + \frac{\partial f}{\partial z} \frac{\partial z}{\partial s} = \frac{\partial f}{\partial x} \frac{\partial g}{\partial s} + \frac{\partial f}{\partial y} \frac{\partial h}{\partial s} + \frac{\partial f}{\partial z} \frac{dk}{ds}.$$

(f) By the Chain Rule

$$\begin{aligned} \frac{\partial u}{\partial s} &= \frac{\partial u}{\partial x} \frac{\partial x}{\partial s} + \frac{\partial u}{\partial y} \frac{\partial y}{\partial s} = \frac{x}{\sqrt{x^2 + y^2}} te^{st} + \frac{y}{\sqrt{x^2 + y^2}} 1 = \frac{te^{2st} + s + \sin t}{\sqrt{e^{2st} + (s + \sin t)^2}}. \\ \frac{\partial u}{\partial t} &= \frac{\partial u}{\partial x} \frac{\partial x}{\partial t} + \frac{\partial u}{\partial y} \frac{\partial y}{\partial t} = \frac{x}{\sqrt{x^2 + y^2}} se^{st} + \frac{y}{\sqrt{x^2 + y^2}} \cos t = \frac{se^{2st} + (s + \sin t) \cos t}{\sqrt{e^{2st} + (s + \sin t)^2}}. \end{aligned}$$

Qu. 5. Find

$$\frac{\partial u}{\partial x}, \quad \frac{\partial u}{\partial y}, \quad \frac{\partial^2 u}{\partial x^2}, \quad \frac{\partial^2 u}{\partial y^2}, \quad \text{and} \quad \frac{\partial^2 u}{\partial x \partial y},$$

where

a) $u = x^y$

b) $u = \ln(x + y^2)$

(Hint: it might be a good idea to check the procedure by calculating the mixed derivatives in two ways, i.e. by differentiating with respect to x first and y second and the other way round.)

Solution. a) If $u = x^y$, then we have

$$\frac{\partial u}{\partial x} = yx^{y-1} \quad (\text{This is like differentiating } x^n)$$

$$\frac{\partial u}{\partial y} = x^y \ln x \quad (\text{You can do this by representing } u = x^y = e^{y \ln x}.)$$

(Eventually, you may even remember that in the A-levels you had $da^x/dx = a^x \ln a$.)

Then

$$\frac{\partial^2 u}{\partial x^2} = y(y-1)x^{y-2}, \quad \frac{\partial^2 u}{\partial y^2} = x^y \ln^2 x.$$

Finally,

$$\frac{\partial^2 u}{\partial x \partial y} = x^{y-1} + yx^{y-1} \ln x.$$

b) Similarly, for $u = \ln(x + y^2)$ we have

$$\frac{\partial u}{\partial x} = \frac{1}{x + y^2}, \quad \frac{\partial u}{\partial y} = \frac{2y}{x + y^2}.$$

For the second derivatives we have

$$\frac{\partial^2 u}{\partial x^2} = -\frac{1}{(x + y^2)^2}, \quad \frac{\partial^2 u}{\partial y^2} = \frac{2}{x + y^2} - \frac{4y^2}{(x + y^2)^2} = \frac{2(x - y^2)}{(x + y^2)^2}, \quad \frac{\partial^2 u}{\partial x \partial y} = -\frac{2y}{(x + y^2)^2}.$$

Qu. 6. In the 2nd-order partial differential equation

$$\frac{\partial^2 z}{\partial x^2} - y \frac{\partial^2 z}{\partial y^2} = \frac{1}{2} \frac{\partial z}{\partial y},$$

change the variables $(x, y) \mapsto (u, v)$, $u = x - 2\sqrt{y}$, $v = x + 2\sqrt{y}$ (implying obviously that $y > 0$).

Solution. Using the Chain Rule, we have

$$\frac{\partial z}{\partial x} = \frac{\partial z}{\partial u} + \frac{\partial z}{\partial v}, \quad \frac{\partial^2 z}{\partial x^2} = \frac{\partial^2 z}{\partial u^2} + 2 \frac{\partial^2 z}{\partial u \partial v} + \frac{\partial^2 z}{\partial v^2}.$$

Then,

$$\frac{\partial z}{\partial y} = -\frac{1}{\sqrt{y}} \frac{\partial z}{\partial u} + \frac{1}{\sqrt{y}} \frac{\partial z}{\partial v},$$

$$\begin{aligned} \frac{\partial^2 z}{\partial y^2} &= \frac{1}{2y\sqrt{y}} \frac{\partial z}{\partial u} - \frac{1}{\sqrt{y}} \frac{\partial^2 z}{\partial u^2} \left(-\frac{1}{\sqrt{y}}\right) - \frac{1}{\sqrt{y}} \frac{\partial^2 z}{\partial v \partial u} \frac{1}{\sqrt{y}} - \frac{1}{2y\sqrt{y}} \frac{\partial z}{\partial v} + \frac{1}{\sqrt{y}} \frac{\partial^2 z}{\partial u \partial v} \left(-\frac{1}{\sqrt{y}}\right) + \frac{1}{\sqrt{y}} \frac{\partial^2 z}{\partial v^2} \frac{1}{\sqrt{y}} \\ &= \frac{1}{y} \frac{\partial^2 z}{\partial u^2} - \frac{2}{y} \frac{\partial^2 z}{\partial u \partial v} + \frac{1}{y} \frac{\partial^2 z}{\partial v^2} + \frac{1}{2y\sqrt{y}} \frac{\partial z}{\partial u} - \frac{1}{2y\sqrt{y}} \frac{\partial z}{\partial v}. \end{aligned}$$

Now, after substituting the above expressions into the original equation, we obtain

$$\frac{\partial^2 z}{\partial u \partial v} = 0,$$

which is the (vaguely) familiar wave equation.

Qu. 7. Find how the equation

$$\frac{\partial^2 w}{\partial x \partial y} + w = 0$$

will be transformed if we look for a particular solution in the form $w = f(u)$, where $u = (x - x_0)(y - y_0)$ (x_0 and y_0 are constants). (In other words, what form will this equation take if we write it down in terms of f and u only?)

Solution. Again, we use the Chain Rule:

$$\frac{\partial w}{\partial y} = \frac{df}{du} \frac{\partial u}{\partial y} = \frac{df}{du} (x - x_0).$$

(Note that here we have df/du , not $\partial f/\partial u$.)

Then,

$$\frac{\partial^2 w}{\partial x \partial y} = \left(\frac{d^2 f}{du^2} \frac{\partial u}{\partial x} \right) (x - x_0) + \frac{df}{du} = \frac{d^2 f}{du^2} \underbrace{(y - y_0)(x - x_0)}_{=u} + \frac{df}{du} = u \frac{d^2 f}{du^2} + \frac{df}{du}.$$

Thus, the equation $\partial^2 w / \partial x \partial y + w = 0$ becomes

$$u \frac{d^2 f}{du^2} + \frac{df}{du} + f = 0,$$

which is now an *ordinary* differential equation, as opposed to the partial differential equation that we had initially. This is because we were looking for a solution that has a particular form.
